Chinmay Murthy

510-358-7145 | cmurthy@uw.edu | linkedin.com/in/chinmaymurthy | gitlab.com/cmurthy | github.com/pie474

Education

University of Washington — Seattle, WA
Bachelor of Science, Electrical and Computer Engineering (3.73, Dean's List) | Music Minor — *Sep 2022 – Jun 2026*
- Relevant Coursework: Learning-Based Control, Machine Learning, Computer Architecture, Embedded Systems

Experience

Robot Controls Software Lead — Sep 2022 – July 2026
Advanced Robotics at the University of Washington (ARUW) — *Seattle, WA*
- Designed and implemented Cascade PID-LQG controller of a 6DOF balancing two-wheel-legged robot (5-bar active suspension), simulated in Matlab/Simulink before STM32 deployment
- Used Reinforcement Learning to train a balancing policy for the same robot with apprenticeship learning
- Implemented inverse kinematics and gravity feedforward for mobile 6DOF differential manipulator
- Designed a Extended Kalman Filter for wheel-CV-fused localization achieving 2-3 cm accuracy over 100m
- Engineered an autonomous path planner/follower that can navigate dynamic obstacles using LiDAR
- Prototyped a Meta-Imitation Learning pipeline in PyTorch and Mujoco distilling RL and MPC controllers into a recurrent foundation policy for online system identification and zero-shot adaptation
- Managed team of 10 to maintain and deploy an open-source 75k+ LoC C/C++ codebase

Software Engineering Intern — June 2025 – Sep 2025
Bidgely — *Palo Alto, CA*
- Performed supervised fine-tuning of a large language model (LLM) on a curated domain-specific dataset
- Designed and implemented a custom retrieval-augmented generation (RAG) pipeline for proprietary data, reducing response latency by 47%

Undergraduate Research Assistant — Oct 2024 – May 2025
University of Washington — *Seattle, WA*
- Optimized real-time vision-SLAM and sensor fusion algorithms (MSCKF-based VINS) leveraging computer vision, IMU, and GPS data for autonomous robot navigation in unmapped environments

Engineering Intern — June 2024 – Sep 2024
Geegah LLC — *Ithaca, NY*
- Brought up and verified RF circuitry in next-gen (4x pixel density) ultrasonic imager with 200+ components
- Reengineered FPGA architecture for 25% lower resource use and 100x performance gain
- Created Python API for USB communication, enabling successful customer demos and IEEE IUS presentation

Applications Intern — June 2023 – Sep 2023
Geegah LLC — *Ithaca, NY*
- Integrated proprietary imager with 3axis motion control stage presented at DARPA ERI Summit 2023

Projects

Optical Triangulation of Ionospheric Events | *OpenCV, Python* — May 2026
- Meteor detection algorithm and trajectory reconstruction pipeline capable of fusing multimodal data. Developed a custom end-to-end simulation environment for accuracy validation

Control Tower (Fleet Management System) | *React, Python, GitLab, sCons* — Apr 2026
- Real-time fleet management dashboard processing 500Hz telemetry and aggregating live network metadata from 20+ distributed Raspberry Pi and Nvidia Jetson compute nodes
- Dynamic IP resolution database and API, integrating with sCons to enable automated firmware flashing to microcontrollers based on real-time hardware target data

Skills

Languages: C/C++, Python, Java, Verilog/SystemVerilog, Matlab, TypeScript
Tools: Git, React, OpenCV, Matlab/Simulink, Unix/Linux, ROS, CAN Bus, I2C, SPI, UART